

Towards Interpretable Machine Learning Models to Aid the Academic Performance of Children and Adolescents with Attention-Deficit/Hyperactivity Disorder

Caroline Jandre[1(✉)], Marcelo Balbino[1], Débora de Miranda[2], Luis Zárate[1], and Cristiane Nobre[1]

[1] Department of Computer Science, Pontifical Catholic University of Minas Gerais, Minas Gerais, Brazil
caroline.jandre@sga.pucminas.br, marcelobalbino@gmail.com, {zarate,nobre}@pucminas.br

[2] Department of Pediatrics, Federal University of Minas Gerais, Minas Gerais, Brazil
debora.m.miranda@gmail.com

Abstract. Educational Data Mining (EDM) integrates numerous auxiliary techniques in capturing, processing, and analyzing school data, with the aim of monitoring and evaluating the process of acquiring knowledge. This assessment and closer monitoring of the student can positively help the learning of students with Attention Deficit Hyperactivity Disorder (ADHD), as they are more likely to have school difficulties, especially in the basic subjects: arithmetic, writing, and reading. Therefore, this work, which is an extension of the article by Jandre et al., seeks to complement the prediction of the results found with the VTJ48 and JRip algorithms that lead to high or low performance of de students whit ADHD in the three basic disciplines, adding the analysis of Random Forest, SVM, and ANN models, in addition to the application of the SHAP method to explain the output of the best model obtained, in case it is not explicitly interpretable. With the results obtained, it can be seen that the best prediction for the arithmetic discipline was performed by Random Forest and SVM (tied); in writing it was the ANN; and in reading it was the VTJ48. In addition, among the features that lead students with ADHD to have a high or low school performance are factors related to parental behavior, student gender, mother's education level, and family financial situation, among others.

Keywords: Interpretability · Machine learning · SHAP · Academic performance · School performance · ADHD · Children · Adolescents

1 Introduction

The search for models for the efficiency and quality of improved teaching and learning stimulated the creation of an area known as Educational Data Mining (EDM), which integrates numerous auxiliary techniques in capturing, processing, and analyzing school data, with the objective of monitoring and evaluating the knowledge acquisition process. All this analysis enables solutions that support the decision-making process in

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educational environments [39], may lead to the construction of a more inclusive environment. This change can positively impact the learning of students with Attention-Deficit/Hyperactivity Disorder (ADHD), as they are more likely to experience school difficulties, disruptive behaviors, and negative interactions with their colleagues, teachers, and parents [33], circumstances that can be softened with directed interventions.

Present in the Diagnostic and Statistical Manual of Mental Disorders (DSM), o ADHD is defined by harmful levels of inattention, disorganization, and/or hyperactivity and impulsivity, which can impact the social, emotional, and academic functioning of the individual, as well as their family [5,33]. Population surveys suggest that ADHD occurs in most cultures in about 5% of children, being more frequent in males. When it occurs in girls, they usually show more characteristics of inattention [5,25].

Since ADHD is not a definitive factor for poor school performance [12], it is essential to identify which other characteristics can enhance or minimize losses in the academic environment. This work is an extension of the article by Jandre et al. [19] that aimed to find classification rules that best describe school performance in arithmetic, writing, and reading (subjects that are part of the first phase of elementary school, provided for in the International Standardized Classification of Education [43]) of students with ADHD, using two classification algorithms: JRip, which generates rules directly from the dataset, and C4.5, a decision tree algorithm.

The objective of this work is to complement the prediction of the results found, adding the analysis of Machine Learning (ML) models: Random Forest (RF), Support Vector Machine (SVM), and Artificial Neural Network (ANN), as these tend to present better performance when compared to other algorithms. In addition, the SHapley Additive exPlanations (SHAP) method will be applied to explain the output of the best model obtained.

As it is an extension, the methodology of the initial stage of data mining, used in this work, followed the one applied in the work of Jandre et al. [19]: we used the same initial database with 266 children and adolescents and 255 features; same pre-processing as the base; and the same three approaches for feature selection: 1) *Genetic Algorithm (GA)*, which works to select the smallest subset of features, representative and relevant, seeking a higher quality model [31]; 2) *RF*, an ensemble algorithm, combining decision trees, which lists the features in order of importance; and 3) feature selection performed by a *specialist in ADHD and literature information* [4].

The article follows the following structure: in Sect. 2, the theoretical foundation is presented, which brings the main concepts related to work. The works related to the theme are covered in Sect. 3. Section 4 presents the methodology used, with a detailed description of the database and the pre-processing steps. Section 5 presents the discussions regarding the results found. Finally, in Sect. 6, the final considerations in this article are exposed.

2 Background

2.1 Attention-Deficit/Hyperactivity Disorder

The term ADHD appeared for the first time in the DSM, in its version III-R, but it was in the DSM-IV that it became known with its three subtypes: *predominantly inattentive*,

when symptoms related to inattention are more accentuated; *predominantly hyperactive/impulsive*, in which the traits linked to hyperactivity and impulsivity are more significant; or *combined*, an association of inattentive, hyperactive and impulsive behaviors, this subtype being what is present in about 62% of people with ADHD [7,25,32].

Is discussed that the cause of ADHD is multifactorial, ranging from biological to environmental aspects, and that the disorder is related to the dysregulation of the neurotransmitters dopamine and noradrenaline. If these neurotransmitters act unsatisfactorily in the frontal lobe and its connections, attention, planning, organization, impulse control, focusing/filtering stimuli, among other functions, are affected, which would explain the symptoms of ADHD [26].

The diagnosis of ADHD is often complex, as there is not a specific test sufficient to define the existence of the disorder, making its identification happen through clinical-behavioral criteria. Furthermore, in 30 to 50% of cases, people with ADHD also have other types of disorders, called comorbidities, which makes the accuracy of symptoms even more ambiguous [23,29].

The treatment for ADHD, it usually requires a comprehensive and multimodal approach, including pharmacotherapy, behavioral measures, and environmental modifications, to reduce behavioral symptoms, as well as providing an improvement in the quality of life of the individual and his/hers relatives [33].

It is noteworthy that the school environment is considered of great help in the cognitive and socio-emotional development of the human being [36]. Thus, the educational institution can help students with ADHD to overcome their academic difficulties, creating strategies that contribute to the inclusion process and ensuring that these students are not at a disadvantage about other people who do not have the disorder [10,26].

2.2 Features Selection with Genetic Algorithm

Two methods stand out for dimensionality reduction in terms of the number of features: 1) *compression of features*, which encodes or transforms the data to obtain a compact representation of the originals, as is the case with Principal Component Analysis; 2) *features selection*, which identifies and discards irrelevant, not very relevant or redundant features. Among the strategies for selecting features, the following stand out: 1) *exponential* (represented by exhaustive search); 2) *sequential* (represented by direct sequential selection); 3) *random* (represented by GA) [31].

Genetic algorithm is a heuristic optimization and search technique based on the principles of genetics and the mechanisms of natural selection [15]. The technique was introduced by Holland [18] and popularized by Goldberg [14] and is based on the Darwinian maxim that *"The better an individual adapts to his environment, the greater his chance of surviving and generating descendants"* [24].

Algorithm 1 is a simplified representation of the GA evolution process. The proposal is to work with a population composed of many individuals to evolve according to a criterion defined by the *fitness* function. The individual is represented by the chromosome, a data structure representing the possible solutions to the problem. The process begins by randomly generating a set of chromosomes, constituting the initial population. This population is evaluated through the fitness function, which determines how well the

individual has adapted to the problem. Next, individuals are submitted to the evolutionary process that involves selecting parents and genetic operators. The selection consists of finding the most suitable individuals (parents) and letting them pass their genes on to the next generation. Once the parents are selected, they are submitted to genetic operators (mutation and crossover) to generate new individuals (children). The new individuals are evaluated by the function *fitness* and passed on to the next generations following the principle of natural selection until the stopping criterion is met.

Algorithm 1. GA Pseudocode.

$t \leftarrow 0$;
$InitializePopulation(P(t))$;
$fitness(P(t))$;
while *not* *Stopping Criteria* **do**
  $parents \leftarrow Selection(P(t))$;
  $children \leftarrow CrossoverMutation(parents)$;
  $P(t) \leftarrow P(t) + children$;
  $fitness(P(t))$;
  $P(t+1) \leftarrow NewPopulation(P(t))$;
  $t \leftarrow t + 1$;
end

2.3 Interpretability

ML systems have become increasingly common in high-risk problems that profoundly impact human life and society. Models have supported issues where a correct decision is essential [38]. Given this, there was a movement in the scientific community searching for increasingly better models from the point of view of predictive performance. Much research has been directed in this direction, and relevant results have been obtained, mainly through the so-called "black-box" methods [1].

However, having a model with high predictive performance is only a partial solution to the problem in many domains. In high-risk scenarios, such as the health area, for example, the decision-maker feels insecure without an explanation of the results of the model [42]. Therefore, it is necessary to present elements that allow developers and users to understand the system's decisions better, increasing confidence in the results and, when necessary, allowing the perception of incorrect decisions.

Faced with the need to explain the results generated by ML models, the scientific community has begun to turn its attention to the design of methods that aim at interpretability. To this end, one can make use of intrinsically interpretable models (e.g., decision tree) or methods that provide *post hoc* explanations for the predictions made by complex models [20].

As illustrated in Fig. 1, interpretability methods present a new perspective for ML solutions by adding an explanation model(g) to the original prediction model(f). The idea is to generate an interpretable approximation of the original model [28].

For this purpose, Lundberg and Lee [27] present SHAP, a tool based on cooperative game theory to interpret ML models. The approach uses *SHAP values* as a way to measure the importance of each feature in a prediction. Thus, each feature impacts a

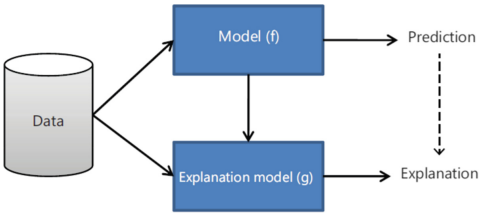


Fig. 1. How an explanation model is used in predicting interpretation [28].

given instance by a value that can be positive, negative, or zero. The authors created a version applicable to any model called *Kernel SHAP* and some specific versions for certain types of models such as *Linear SHAP*, *Low-Order SHAP*, *Max SHAP*, *Deep SHAP* and the *SHAP Tree Explainer*.

3 Related Works

The literature has been progressing about research focused on EDM, with works aimed at acquiring knowledge on educational topics, such as student performance prediction [2], learning through remote teaching [16], among other issues related to the school environment.

The work of Wandera et al. [44] revealed characteristics associated with good academic performance in high schools in South Africa, using Logistic Regression, Light Gradient Boosting (LightGBM), and tree-based algorithms. The database used combined sources related to community research, students' school performance, among other information. The SHAP and odds ratios were used to explain the model results and provide information on the relationships between the variables. The Logistic Regression and LightGBM models outperformed the decision tree model, and the significant characteristics presented by SHAP indicate that the availability of potable water in the community, the situation of the schools' bathrooms, the possibility of using cellular internet by students, among other variables, impact the school performance of students.

Souza and dos Santos [39] used a database that addresses the performance in Mathematics and Portuguese of students in secondary education in two Portuguese schools. The objective was to compare consolidated techniques within the scope of the EMD, with the Deep Learning technique. The ML algorithms chosen were: Naïve Bayes, Decision Tree, RF, and SVM. For Deep Learning, the MultilayerPerceptron (MLP) architecture was used. The authors point out that the Decision Tree was the second most accurate model, reaching 87% accuracy, and that, due to its simplicity of application and processing speed, it can be a promising alternative. Regarding the prediction of school performance, the authors found the grades and the number of student absences as the main features. However, they emphasize that the influence of other elements cannot be ruled out, such as the experience of adversities by the students, but that, due to the lack of information in the questionnaires that evidence what motivated the students' grades and attendance, it is only possible to say that the influential features are those related to their school performance.

The literature also reveals that ML methods have been widely used in research related to ADHD and have generated important advances in the search for knowledge associated with this disorder.

Anuradha et al. [3] conducted a study for diagnosing ADHD in 100 children aged 6 to 11 years using the SVM algorithm. The database was collected through the students' responses to a questionnaire and medical diagnoses indicating whether the children had ADHD. The work used GA to features selection in order to identify the essential features and increase the accuracy of the model. The study achieved an 88.7% success in diagnosing ADHD, which was considered a satisfactory result from the authors.

Rahadian et al. [35] used GA to improve a Learning Vector Quantization 2 Neural Network (LVQ2NN) method to classify data about the type of ADHD in patients. The GA was used to optimize the weight vector in the training process. The tests performed without the GA reached 80% of correctness, while the model with the GA improved the performance to 89.5%.

Therefore, this work is in line with the literature, as it uses algorithms frequently used in other articles, and techniques for interpretability and selection of known features. However, there is still not a wide range of studies relating data on the lives of people with ADHD and EDM. Thus, this extension of the article by Jandre et al. [19] may contribute to more findings focused on this area of interest.

4 Materials and Methods

4.1 Description of the Database

The database was made available by the Department of Pediatrics of the Federal University of Minas Gerais - Brazil. The sample consists of 266 students (instances), aged between 6 and 18 years. Among them, 196 are diagnosed with ADHD and 70 have a negative diagnosis.

The database, based on 225 characteristics (features), it was built from the individual and family responses present in the questionnaires. Also, interviews were conducted with those responsible and patients who are followed up at the hospital linked to the college. The base contains information on health, financial conditions, parental care, education, among others, in addition to the notes for the arithmetic, writing, and reading tests of each present in the database.

4.2 Pre-processing

With the aim to improve the quality of the data, the pre-processing of the database was performed through the steps described below.

1. Exclusion of instances that did not present the grade of the Test of School Performance (TSP) [40] in arithmetic, writing, or reading.
2. Transformation of the grades obtained in the TSP, comparing them with the average grade of a Brazilian state, in High and Low performance. For this, the Standards Tables present in the TSP manual were used. The tables in the manual for the classification of notes range from the 1st to the 6th grade, which corresponds from the

2nd to the 7th year in the current academic category. Therefore, the years before the 2nd year were classified according to the classification criteria of the 2nd year, and those after the 7th year followed the classification parameters of the 7th year.
3. Exclusion of irrelevant features (e.g., name and telephone number), and those who presented the same information or were complementary (e.g., age in months and age in years), concatenating them and maintaining only one feature.
4. Transformation of features belonging to the same category into a single feature (e.g., Conduct Disorder and Opposition Disorder are part of Behavior Disorders, so a single feature called "Behavior Disorders" has been created, which indicates whether the individual has "Conduct Disorder" or "Opposition Disorder").
5. Filling in the missing data by the average, in numerical data, or by mode, in categorical features.
6. Binarization of non-ordinal nominal features, that is, they were coded as the presence or absence of the characteristic.
7. Random manual separation of 15% of each class's instances to carry out the tests.
8. Balancing of 85% of the remaining data using the SpreadSubsample algorithm present in the Waikato Environment for Knowledge Analysis - WEKA[1] The SpreadSubsample algorithm follows the undersampling approach, randomly reducing instances of the majority class. Uniform distribution was applied. After balancing, the Randomize filter was executed in order to shuffle the instances.

Thus, Table 1 shows the total number of instances, of each class, for testing and creating models, after the pre-processing steps. It should be noted that the separation of the instances for testing was carried out before the database was balanced.

Table 1. Number of balanced instances for creating the model and unbalanced for the tests [19].

Discipline	Training/Validation		Test	
	High	Low	High	Low
Arithmetic	59	59	11	29
Writing	50	50	09	31
Reading	39	39	08	27

As for the features, at the end of the pre-processing, 130 remained to represent the base. Figure 2 presents an overview of the database, with the categories of data and the number of features in each category, in the form "Number of initial features/Number of features after pre-processing".

4.3 Dimensionality Reduction with Genetic Algorithm, Random Forest, and Specialist

A combination of three methods - GA, RF, and specialist - was used to select the essential features in learning among the 130 features present in the database.

[1] WEKA is open source software issued under the GNU General Public License that contains a collection of ML algorithms [13]. Available at http://www.cs.waikato.ac.nz/ml/weka.

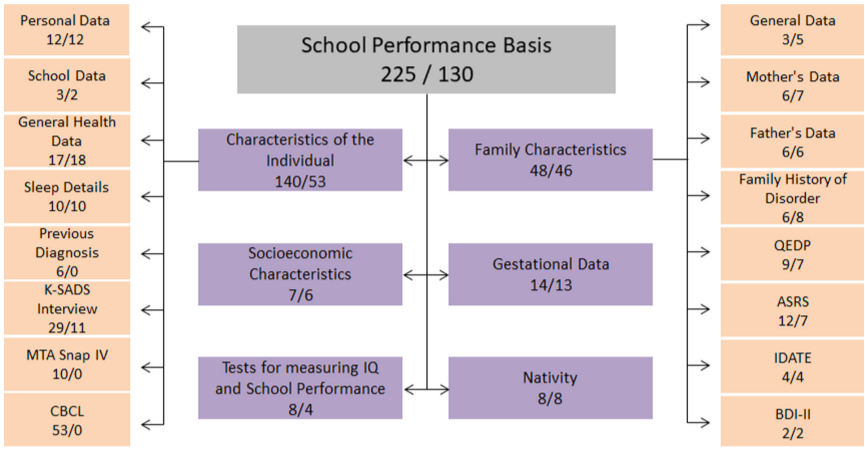


Fig. 2. Overview of the database, grouped by categories [19].

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) algorithm was chosen to find the best subset of features maximizing your fitness, in this case, the F-Measure. To evaluate the F-Measure, the K-Nearest Neighbors (KNN) classifier was used. His choice was motivated by the low computational cost for adjusting his parameters. In this work, the parameter k (number of closest neighbors) was adjusted, varying its value range [1–10]. GA was implemented in the Python language, using the DEAP library, available from Université Laval [11].

Aiming at the adequate definition of the necessary parameters for the experiments' execution, intervals of values were analyzed, having as a stopping criterion the number of GA generations. For each set of parameters, 10 different random seeds were used. Table 2 shows the ranges of values used.

Table 2. Range of parameter values [19].

Population initialization	Random
Representation	Binary
Crossover operator	Two Points
Crossover probability	70%, 75%, 80% and 85%
Mutation operator	One Point
Mutation probability	1%, 5% and 10%
Population size	100, 300 and 500
Number of generations	100, 300 and 500
Crossover selection method	Tournament ($size = 2$)
Composition of the new generation	Non-dominated individuals
Stopping criteria	Number of generations

To automatically adjust the RF algorithm's parameters, CVParameterSelection[2] was used [21], varying the number of trees and features, and the depth of the trees. The RF calculates the feature's importance, indicating which are the most important for creating the model. Only features with relevance as from 70% were considered.

Regarding specialist, the knowledge of the medical expert in ADHD, who provided the database and is a collaborator in this work, was used to finalize the features selection, in conjunction with the information present in the work of [4], focused on the school performance.

Figure 3 shows the features selected by GA and RF, separated for each of the three disciplines (Figs. 3a, b and c). The figure also shows the features considered important, for the three disciplines (Fig. 3d), according to the specialist.

4.4 Description of Methods

To assist in discovering the rules that lead the student to obtain a high or low performance, the VTJ48 algorithm (tree algorithm) and the JRip (rule algorithm) were used. VTJ48 [41] has the same functioning as J48, developed by Quinlan [34], which is a Java adaptation of C4.5. That is, the VTJ48 also builds a decision tree from the instances, but its difference is that it automatically adjusts the confidence for pruning and the minimum number of instances per leaf. JRip is a Java implementation of the Ripper algorithm, proposed by [9] as an optimized version of the IREP algorithm. JRip constructs the rules seeking to represent the model as compactly as possible with as much information of the data; that is, it seeks to consistently explain which features are relevant to the pattern(s) found in the database, with the minimum of rules.

Regarding the "black-box" algorithms, RF, MLP, and Sequential Minimal Optimization (SMO) were used. WEKA's RF is an extension and improvement of bagging for decision trees, and can be used for classification or regression [6]. RF uses the voting system, combining the response of several trees, to determine the class of an instance. MLP is an implementation of ANN. This algorithm follows a flow that starts at the input layer and goes through n layers, until reaching the output layer (which represents a given problem), using backpropagation to classify the instances [30]. The SMO is an implementation to train an SVM classifier, which aims to find the hyperplane that presents the greatest distance between the existing classes in the database, maximizing its separation [17].

Two strategies were used to deepen the understanding of the model and consequently the comprehension of the features that impact the performance in arithmetic, reading, and writing: 1) if the best predictive model of performance is the decision tree, which is intrinsically interpretable, we use the rules obtained through the tree; 2) if the best model is based on one of the "black-box" algorithms, we propose to perform the interpretability using the SHAP approach.

About SHAP, as shown in Sect. 2.3, the tool generates an explanation model from the classification model. The explanation model takes as input a sample of data and performs interpretability for the predictions of this sample. We chose to apply the explanation model to the same set of instances in which the classification model tests were

[2] Available in the WEKA tool.



Fig. 3. Description of the features of the three databases, considering the three selection methods [19].

applied. The graphs generated by SHAP allow us to identify and understand the most relevant features for the predictions.

4.5 Model Quality Assessment Metrics

To evaluate the quality of the obtained models, Precision, Recall, and F-Measure metrics were used.

Precision[3] is the percentage of instances correctly classified in a class, out of all those that were classified in the class.

Recall[4] is the percentage of instances of a class that have been correctly predicted to belong to the class.

F-Measure[5] represents the harmonic mean between Precision and Recall.

All classifiers were built and validated using the k-fold cross-validation process, with k = 10 [22].

5 Results and Discussions

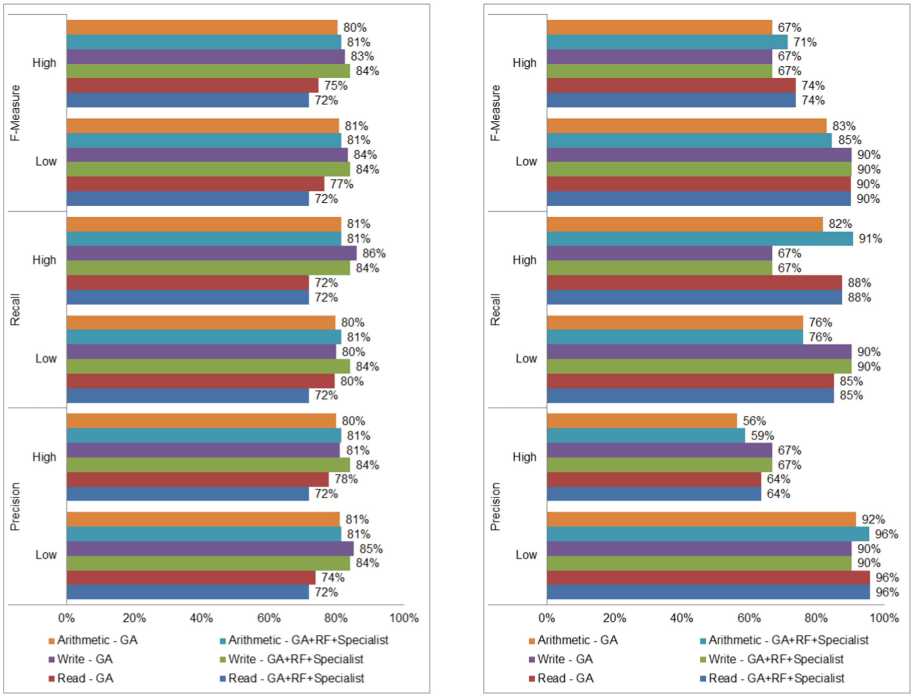
Figures 4, 5, 6, 7, and 8 present the results of the model creation and testing phase of the RF, MLP, SMO algorithms, JRip, and VTJ48, respectively. The values refer to the experiments with the selection of the GA and the insertion of the characteristics of the RF and specialist. The results are presented by discipline and by the algorithm. We chose to make a graph for each algorithm to highlight their behavior in each experiment separately. It is noteworthy that in the experiments in which the features were selected only by the GA, the number of features used in the prediction of arithmetic, writing, and reading performance was only 16, 29, and 7, respectively. In the experiments in which the features are a combination of the non-identical ones present in the selection of the GA, RF, and Specialist, 51 features remained for arithmetic representation, 54 for writing, and 43 for reading.

Analyzing the basis used for the test and observing the F-Measure metric, it can be seen that the RF algorithm, Fig. 4, obtained better results: with the features selected by GA, RF, and Arithmetic Specialist; and that in writing and reading, the two versions of the base achieved the same result.

The MLP, observing Fig. 5 and following the same analysis parameter, also stood out, in arithmetic, with the GA, RF, and Specialist versions, especially when compared to the high class, which is the original minority class, and, consequently, the one with the greatest difficulty for prediction. In the writing discipline, the best results were also obtained by the combination of GA, RF, and Specialist. However, in reading, the version with features only selected by the GA obtained the best results.

Analyzing the SMO algorithm (Fig. 6), again the version of GA, RF, and Specialist had the best result in arithmetic. In writing, there was no difference between the versions. In reading, the best result was obtained with the features selected by the GA.

[3] $Precision = \frac{VP}{VP+FP}$.
[4] $Recall = \frac{VP}{VP+FN}$.
[5] $F - Measure = \frac{2 \times Recall \times Precision}{Recall+Precision}$.



(a) Training/Validation (b) Test

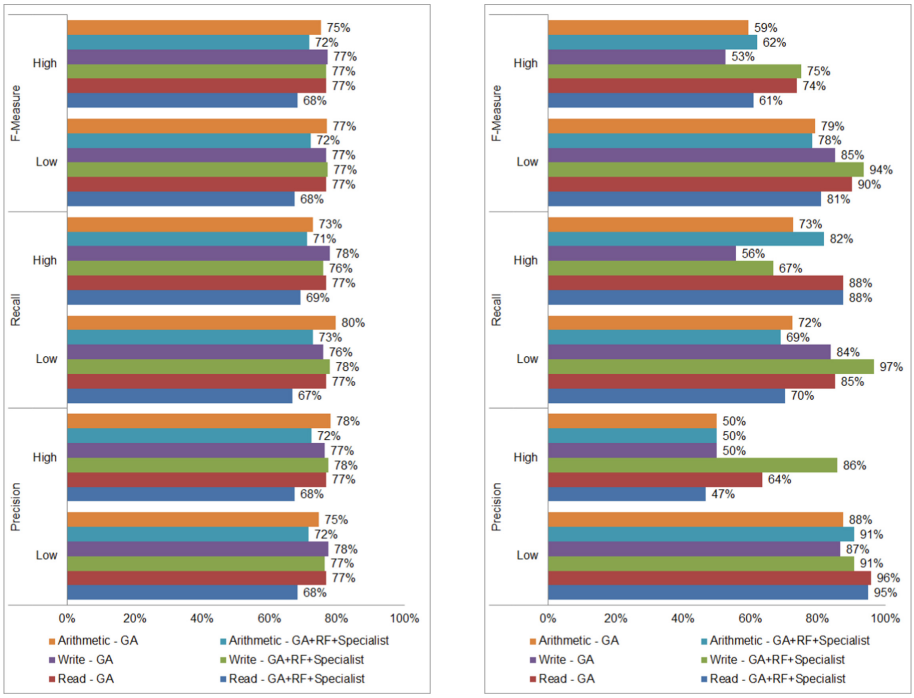
Fig. 4. RF algorithm result.

There was a tie between the two versions of the base in arithmetic, when analyzing the JRip algorithm (Fig. 7). Writing excelled with the version of features selected by GA, and reading with those selected by GA, RF, and Specialist.

Finally, in the VTJ48 algorithm, observing Fig. 8, it can be seen that in arithmetic the version with the features selected by AG, RF, and Specialist was the one with the best result. In writing and reading, it was the GA-only version that stood out.

After this investigation, it is noted that: in arithmetic, the version of the base with the features selected by the GA, RF, and Specialist was the one with the best use, as it presented the best result in 4 of 5 evaluated algorithms, and tied for 1, that is, it was not the worst result at any time; in writing, the best version depends on the algorithm applied, as there was a tie in results in 2 algorithms, a superior result with GA, RF, and Specialist in 1, and only with the features selected by the GA in 2 algorithms. In reading, it can be said that the best version is the one with the features selected only with GA, as it presented a superior result in 3 algorithms, tied in 1, and was inferior in 1 algorithm.

To identify which algorithm had the best prediction in each of the disciplines, another comparison was performed, analyzing only the F-Measure metric, in the test phase, with the bases of the best version previously reported in the description of the



(a) Training/Validation (b) Test

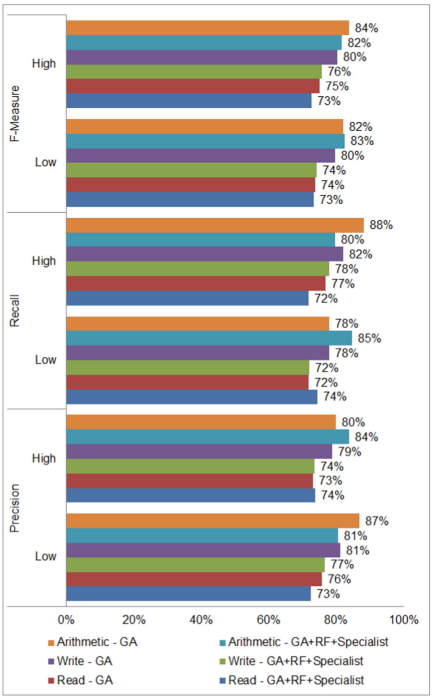
Fig. 5. MLP algorithm result.

results of each of the algorithms (during the analysis of Figs. 4, 5, 6, 7, and 8). Figure 9 presents the new comparison.

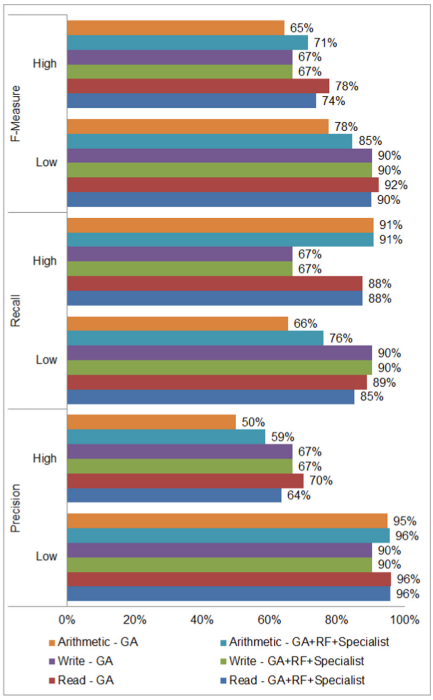
After analyzing Fig. 9, it is noted that, in reading, the algorithm with the highest performance was VTJ48, which is a Decision Tree algorithm; in writing it was the MLP, which is an ANN implementation; and in arithmetic there was a tie between the RF (ensemble algorithm) and the SMO (implementation to train an SVM classifier).

Note that to find the best results described, the following settings were used:

- **VTJ48, in the reading base, with features selected only by the GA:** minimum number of instances per sheet = 2; confidence limit for pruning = 0.5.
- **MLP, in the base of writing, with features selected by GA, RF, and Specialist:** learning rate = 0.4; momentum = 0.3; number of epochs = 600; number of hidden layers = 39.
- **RF, in the arithmetic base, with features selected by GA, RF, and Specialist:** number of trees = 100; number of features = 8; maximum depth of trees = 4.
- **SMO, in the arithmetic base, with features selected by GA, RF, and Specialist:** complexity constant = 2; kernel = puk.



(a) Training/Validation



(b) Test

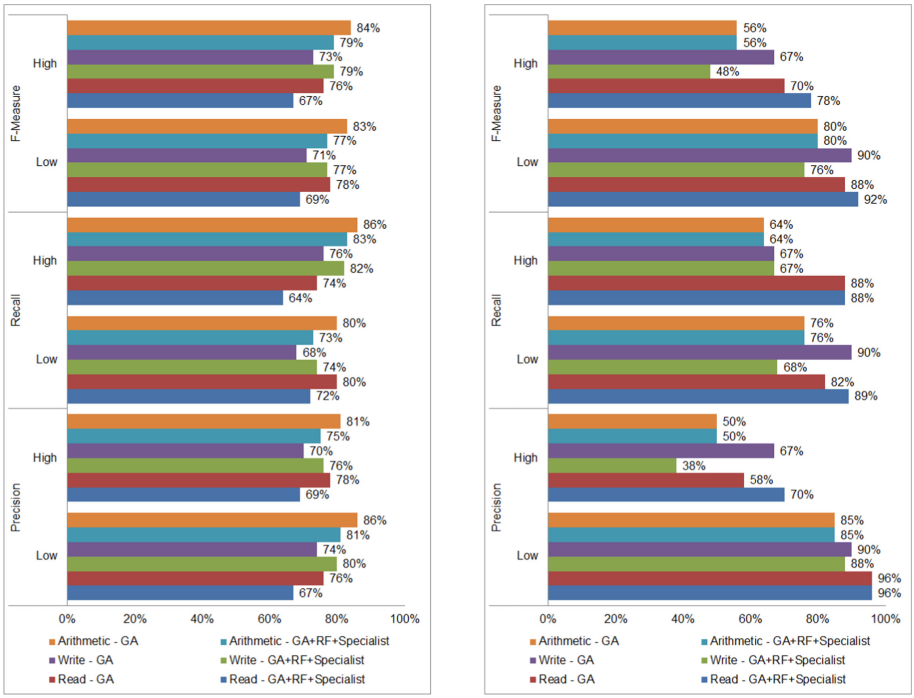
Fig. 6. SMO algorithm result.

5.1 VTJ48 Rules and Interpretability Using SHAP

In the experiments, with the reading base, the algorithm that had the best performance was the VTJ48, which already allows exploring its rules explicitly. Therefore, the Table 3 Table presents the rules found that explain the reasons for a student to have a high or low performance in the reading discipline, with coverage from 10%. Coverage is the percentage of instances that the rule classifies correctly, out of all instances of the class.

Table 3. Generated rules and coverage [19].

Read		
Number	Rule	Coverage
1	If writing performance = low and indulgence $\leq$ 15 then **Low**	80%
2	If writing performance = high then **High**	44%
3	If writing performance = average and fear of sleeping alone = no then **High**	26%

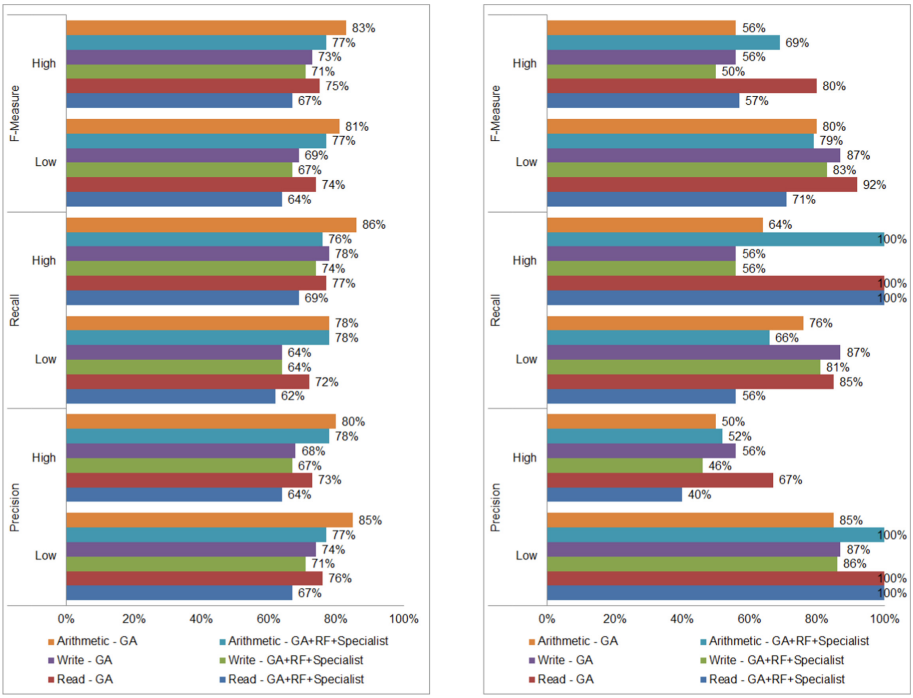


(a) Training/Validation (b) Test

Fig. 7. JRip algorithm result.

Analyzing the rules found in reading, it is noted that the performance in writing has a strong influence on the student's performance in this discipline, especially in rule 2, where if the student does well in writing, he also does well in reading. However, the influence of some other features on reading performance is also observed. Rule 1, for example, with coverage of 80%, points to the importance of the parental behavior, because if the student has a low writing performance and his parents are less indulgent (the highest present value in the base is 24), that is, generally do not forgive mistakes or praise successes, their reading performance will also below. This observation indicates that family support is essential, especially in times of difficulty. Rule 3 suggests that when writing performance is average, and the student is not afraid to sleep alone, he has a high reading performance. One of the justifications for fear of sleeping alone may be linked to phobic anxiety, which causes a lack of confidence, security, and support, leading to feelings of helplessness, fragility, and dependence [37]. Therefore, not being afraid to sleep alone can represent certain self-confidence, which results in a positive performance.

On the other hand, in arithmetic and writing, the best performance was obtained through "black-box" algorithms, which implies restrictions on the interpretability of model decisions. In several problems, as is the case in the present research, limitations in the interpretability of decisions undermine the confidence in the model and its consequent use in practice. Furthermore, the opportunity for knowledge discovery is also reduced.



(a) Training/Validation (b) Test

Fig. 8. VTJ48 algorithm result.

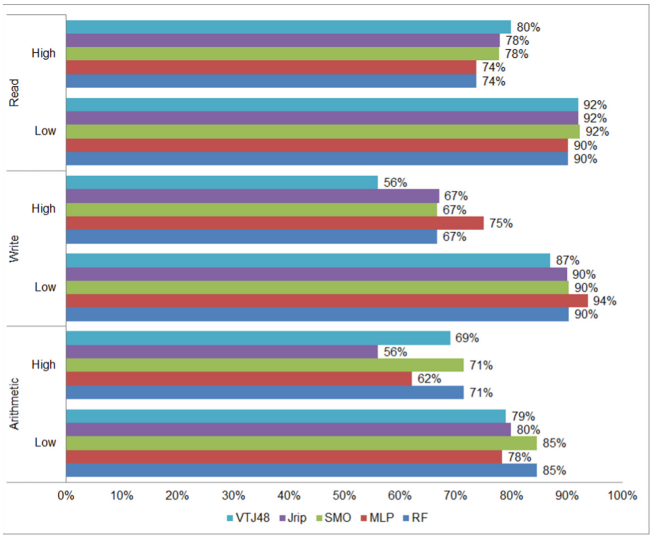


Fig. 9. Comparison between algorithms.



Fig. 10. Impact of features on the arithmetic model.

Thus, we chose to use SHAP to deepen our understanding of model decisions and identify the features with the most significant impact on predicting performance in arithmetic and writing. Figure 10 exemplifies one of the graphs from the tool. It presents an overview of the effects of the main features on predicting performance in arithmetic.

The graph sorts the features by their importance to the model. Each point present on the line of a particular features represents an instance and, consequently, a student that was impacted by that features. If a point is on the right side relative to the central axis, such a feature moves the student closer to the low class. The dot color represents the feature's value in the instance (red for higher values, blue for lower values). It is also important to note that the further away from the graph's central axis, the more significant the impact of that feature for a given instance. Note that the features at the bottom of the graph have a few points closer to the central axis, which means that this feature has a little significant effect on the model.

We see from the graph in Fig. 10 that performance in writing is the feature with the most significant impact on prediction in arithmetic. More specifically, students with lower performance in writing tend to have lower performance in arithmetic. Also, students with high or intermediate performance in writing tend to have high performance in arithmetic.

The graph shows the importance of the mother's schooling, with a positive and significant influence on mothers with completed graduation. We assume that this feature may be related to the support that mothers can offer to their children's learning. Similarly, the family's social class is highlighted, possibly associated with the resources provided to students that require financial support, especially in the case of students with ADHD.

Furthermore, the graph indicates the importance of "Age" on the features with the most significant impact. To delve deeper into the behavior of a particular feature, SHAP has the graph present in Fig. 11. In this case, we can observe the effect of age on predicting performance in arithmetic. It is possible to observe a trend of more outstanding academic difficulty of the student in this subject with the advancement of school years, especially from the age of nine (positive SHAP *values* for the Lower class). We believe that this influence of age is related to the fact that younger children are beginning school life and thus are little impacted by academic activities. As years go by, the difficulty level of the activities increases, which can cause learning problems and, consequently, lead to a worsening in academic performance.

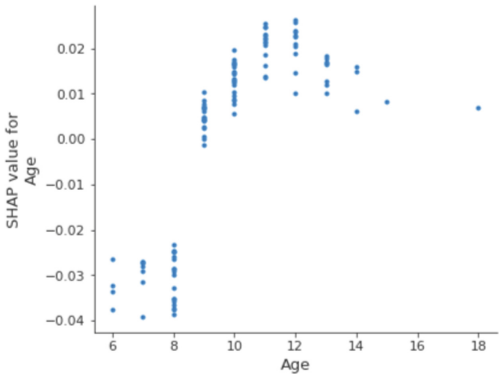


Fig. 11. Impact of age on arithmetic.

Other features are listed but with less importance. Among these, the features "ADHD" and "combined ADHD" stand out, indicating that although ADHD impacts academic performance in arithmetic, other features have a more significant influence. It is noteworthy that over the years, the characteristics of ADHD may have become more evident, which is probably why this feature is not as prominent in this analysis since there are children of early school age.

About the writing model, Fig. 12 shows the list of the most important prediction features in this discipline. There is a relationship between performance in writing and the other disciplines present in the database, so students with lower performance in arithmetic and reading also tend to have low performance in writing and vice-versa.

Next, the graph shows the importance of the feature ADHD, in which having ADHD is a complicating factor for performance in writing. The gender feature also stands out, indicating that being male makes the prediction of lower performance closer while being female contributes to higher performance. The fact that the male gender tends to low performance may indicate a reproduction of social stereotypes in which girls are said to be quieter, and boys are rowdy and undisciplined, which leads to a negative evaluation of the child's behavior and influences the final grade [8]. Other features complete the list but with less importance.

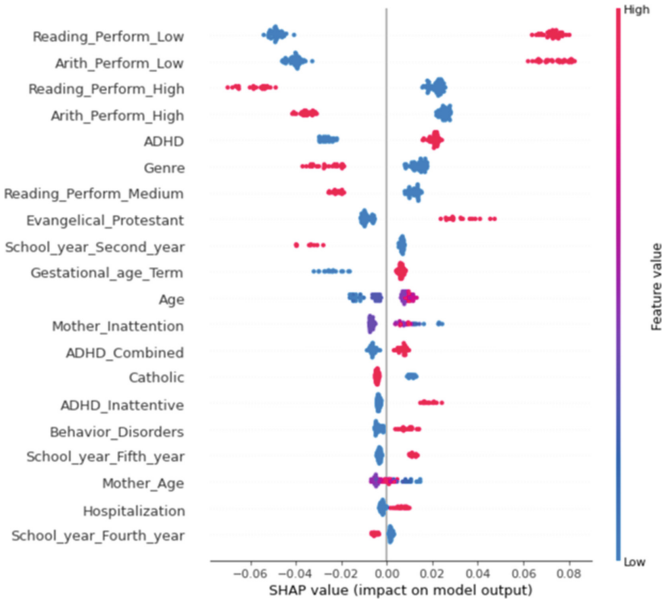


Fig. 12. Impact of features on the writing model.

After analysis, it is clear that performance in students can be affected by factors related to parental behavior, circumstances associated with fear and insecurity (sleeping alone), mother's education, family's financial conditions, gender, age, and student's height, in addition to their performance in other disciplines. ADHD also impacts performance, but generally less so than other aspects of an individual's life. Therefore, a multifactorial approach must be taken so that students with ADHD can improve their academic performance.

6 Final Considerations

As much as society is paying more attention to issues related to ADHD, there is still little knowledge disseminated to the population about the characteristics and implications of the disorder. The judgments that children and adolescents with ADHD are normally affected by can lead to the exclusion of these people, resulting in low self-esteem and anti-sociability. In addition, if the student with ADHD receives due attention from parents and school members, it is possible to change the prognosis considered unfavorable. Therefore, it becomes significant to identify which characteristics influence the individual's academic experience to alleviate their difficulties.

In this work, three algorithms considered "black-box" (RF, MLP, and SMO) and two considered "white-box" (VTJ48, and JRip) were used. With the results, it is noticed that explicitly interpretable algorithms do not always lose in performance, because, in the reading base, it was the VTJ48 that obtained the best prediction result. Therefore, the use of the "white-box" method proved to be positive in this context. In arithmetic and

writing, the association of the "black-box" and SHAP algorithms was efficient, as it was possible to achieve good predictions and, later, identify the most significant features for academic performance, deepening the understanding of the model decisions and contributing to increasing knowledge about the problem in question. According to the observations made, some factors are more relevant than whether or not the student has ADHD. This is an indication that the disorder itself does not indicate how the student's academic performance will be, as the environmental and emotional factors in which he is involved directly affect his attainment of academic knowledge.

Regarding dimensionality reduction, the two versions of the database (one only with features selected by GA, and another with features selected by GA, RF, and Specialist), contributed to a good predictive result, with each version having its own contribution more directed to one or another discipline.

In future work, to identify other factors that may affect the performance of students with ADHD, we intend to evaluate new approaches for balancing, since the use of different instances can contribute to new discoveries. In addition, it is considered to investigate what the results would be like if the problem were treated as multi-label. In other words, it is intended to evaluate the quality of the models in the prediction of student performance in all subjects, jointly.

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